

Deepak K

EEE Undergraduate — Robotics, Embedded Systems & Autonomous Navigation

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EDUCATION

New Horizon College of Engineering

B.E. in Electrical and Electronics Engineering | CGPA: 9.13/10.0 (through 3rd Year)

- Relevant Coursework: Embedded Systems, Control Systems, Microcontrollers, Electronics Circuits, VLSI Design

Bengaluru, Karnataka

Aug 2023 – May 2027

St. Francis Composite PU College

Pre-University Course (PCME) | 92.17%

Bengaluru, Karnataka

Graduated 2022

TECHNICAL SKILLS

Programming: Arduino C, C, Python

Robotics & ROS 2: ROS 2 (Jazzy), micro-ROS, Navigation2, SLAM Toolbox, AMCL, URDF, Behavior Trees, TF Transforms

Autonomous Systems: LiDAR SLAM, Monte Carlo Localization, Sensor Fusion (LiDAR + IMU), Path Planning

Computer Vision & AI: YOLOv8, OpenCV, Real-Time Object Detection, Deep Learning, Transfer Learning, Reinforcement Learning

Control & Embedded: PID, MPC, Inverse/Forward Kinematics, ESP32, Arduino, Raspberry Pi 4, Real-Time Systems

Hardware: 2D LiDAR, IMU, Encoders, L298N Motor Driver, IR Sensor Arrays, Ultrasonic Sensors

Tools: Gazebo, RViz, Git/GitHub, Linux (Ubuntu), Google Colab, MATLAB, Xilinx Vivado, KiCad, EasyEDA

TECHNICAL PROJECTS

Hexapod-6: Autonomous ROS 2 Legged Robot | ROS 2, NVIDIA Jetson, STM32, Python, RPLIDAR

2026

- Engineered a six-legged autonomous robot using ROS 2, with Jetson for high-level compute and STM32 for real-time servo control
- Designed a coordinated six-leg gait control scheme with IMU-based orientation feedback to maintain stable locomotion over uneven terrain, paired with LiDAR SLAM for concurrent environmental mapping
- Integrated RPLIDAR, MPU6050, and STS3215 smart servos via Jetson–STM32 serial link for coordinated sensing and motion control

Autonomous SLAM Robot with Real-Time Monocular Depth Vision | ROS 2, TurtleBot, Raspberry Pi 4, Depth Anything V2, GPU

- Built an autonomous ROS 2 TurtleBot with SLAM Toolbox for real-world mapping and navigation
- Implemented real-time monocular depth estimation using Depth Anything V2, offloading inference to a GPU host while the Raspberry Pi handled sensor capture and control, achieving ~40 FPS end-to-end
- Diagnosed a dead onboard WiFi module blocking remote development; isolated the fault via systematic elimination and migrated to Ethernet to restore SSH access

ROS 2 MPC Navigation System | ROS 2, Python, Gazebo, SciPy

2026

- Implemented nonlinear MPC path tracking for TurtleBot3 in ROS 2 Jazzy over a 15-step horizon using SciPy SLSQP with warm starting for real-time execution
- Built cubic spline path smoothing from CSV waypoints and integrated LiDAR soft-barrier obstacle avoidance into the MPC cost function, enabling the robot to slow and reroute around obstacles without the abrupt stops seen with Pure Pursuit

Soil Grain Detection & Mapping System | YOLOv8, OpenCV, Raspberry Pi, Folium

2025

- Developed a YOLOv8-based soil grain classification pipeline on a custom dataset, achieving 92% detection accuracy across soil categories
- Deployed the trained model for real-time Python/OpenCV inference on Raspberry Pi hardware, with GPS-tagged detections rendered as an interactive map via Folium

EXPERIENCE & LEADERSHIP

Project Committee Member — Embedded Systems Club

New Horizon College of Engineering

Aug 2024 – Sep 2025

Bengaluru, Karnataka

- Organized workshops on embedded systems for student communities, supporting hands-on learning around microcontrollers
- Collaborated on club projects involving sensor integration, circuit design, and embedded programming

CERTIFICATIONS

- Getting Started with AI on Jetson Nano — NVIDIA Deep Learning Institute
- UR20/30 Industrial Robot e-Learning — Universal Robots Academy